SALMAN ALHARBI

Data Science & AI Engineer | ETL Pipelines, Sovereign Local LLMs & Computer Vision

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EXECUTIVE SUMMARY

Data Scientist & AI Engineer specializing in high-throughput Python/SQL ETL pipelines, Sovereign Local LLMs (Ollama, DeepSeek, RAG), and Deep Learning Computer Vision (YOLOv8, EfficientNetB0). Delivered production systems across SERA, Tuwaiq Academy, and NCM harvesting 500K+ regulatory records in 15 minutes, automating meteorological workflows with Delta Lake & DuckDB, and authoring a 6-page IEEE research paper on a 3-model insurance claims pipeline trained on 23,874 images.

PROFESSIONAL EXPERIENCE

Tuwaiq Academy (in partnership with Ministry of Education) - Jeddah, Saudi Arabia **07/2026 - 08/2026**

Lead Technical Instructor (Technical Trainer) | *Awarded Executive Leadership Commendation*

- **Regional Cohort Leadership:** Led gifted, advanced technologies, and AI tracks at Jeddah hub, serving as primary liaison with academy leadership.
- **Technical Translation:** Delivered university-grade Generative AI, MLOps, and Cybersecurity modules; guided IoT (Arduino) & VR (Unity) prototyping.
- **Executive Event Direction:** Orchestrated final grand tech exhibition and coached student presenters, earning formal leadership commendation.

Saudi Electricity Regulatory Authority (SERA via UNIVERSAL STEPS GROUP) - Riyadh, KSA **12/2025 - 06/2026**

Data Science & Decision Intelligence Specialist (Tamheer Certified)

- **Big Data Scalability:** Engineered ETL ingestion pilot harvesting 4-year data backlog (500K+ records) in 15 mins; scaled to 4 additional enterprise clients.
- **PDPL Sovereign AI:** Compressed decision cycle from hours to <5 seconds via Local LLMs, ensuring 100% in-Kingdom residency per PDPL & NDMO.
- **Executive Intelligence:** Automated operational bottlenecks and multi-platform sentiment analytics (5+ channels) into executive oversight dashboards.

National Center of Meteorology (NCM) - Jeddah, Saudi Arabia **06/2024 - 08/2024**

Data Engineer Intern

- Architected automated Python ETL pipelines for hourly weather reports, cutting latency from hours to seconds.
- Integrated Delta Lake for transactional reliability and optimized SQL queries in DuckDB using Docker and Git.
- Eliminated repetitive manual data entry, freeing operational bandwidth for a 15-member analytics team.

EDUCATION & ACADEMIC HONORS

Bachelor of Science in Data Science **08/2020 - 01/2025**

College of Computer Science and Engineering | *University of Jeddah*

- **ABET Honors:** Selected by faculty to represent the Data Science cohort in live interviews before the international ABET board.

FEATURED PROJECTS & RESEARCH

Automated Insurance Decision Support System | Principal Investigator & First Author (6-Page IEEE Paper) **10/2024 - 01/2025**

- Engineered 3-model pipeline (YOLOv8 & EfficientNetB0) across 23,874 images: 81% detection, 93% severity, 100% part ID (21 parts).
- Built rule-based actuarial SAR cost estimator and Streamlit decision platform exporting audit-ready PDF claims dossiers in <60s.

Saudi Real Estate Investment Valuation Engine | Predictive Analytics **10/2023 - 11/2023**

- Achieved R-squared = 0.919 with optimized RandomForest Regressor after benchmarking 7 ML architectures on Saudi regional market data.

CERTIFICATIONS & CREDENTIALS (19)

- Google Cloud Generative AI Leader Track (5 Courses) **DataCamp**
- Gen AI Apps: Transform Your Work & Navigate Landscape **Google Cloud**
- Fundamentals of Artificial Intelligence | SDAIA **09/2025**
- Linear Algebra & Calculus for ML | DeepLearning.AI **11/2023**
- Gen AI Agents: Transform Your Organization **Google Cloud**
- Gen AI: Unlock Foundational Concepts & Beyond Chatbot **Google Cloud**
- Google Data Analytics Professional Cert (8 Courses) **04/2023**
- Python Basics (Michigan) & MS 365 Specialization **07/2025**

CORE COMPETENCIES & TECHNICAL STACK

- **Governance & Advisory:** Saudi PDPL, NDMO Frameworks, Data Sovereignty, Privacy-by-Design, Decision Intelligence, ROI Analysis, Technical Translation.
- **AI & Data Engineering:** Python, SQL, DuckDB, Delta Lake, Local LLMs (Ollama, DeepSeek), RAG, Agentic Workflows, YOLOv8, EfficientNetB0, OpenCV.
- **BI, DevOps & Languages:** Power BI, Tableau, SAP, Streamlit, Pydantic, Docker, Git, R | Languages: Arabic (Native), English (Fluent).